

# ResearchPilot AI - Technical Report

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Vector Databases: A Comprehensive Review

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## ***Executive Summary***

Vector databases are a new type of database designed to handle high-dimensional vector data, which is essential for various applications, including semantic search, recommendations, and natural language processing. This report provides a comprehensive review of vector databases, including their core functionalities, architectural differences, indexing techniques, and best practices. We also discuss the strengths and weaknesses of different vector databases, such as ChromaDB, Pinecone, and FAISS, and provide guidance on how to choose the right vector database for specific use cases.

## ***Key Findings***

- Vector databases are designed to store and query high-dimensional vector embeddings, which are dense numeric representations produced by machine learning (ML) models.
- The core functionalities of vector databases include vector storage, similarity search, and indexing.
- Different vector databases have distinct architectural differences, such as ChromaDB's graph-based indexing, Pinecone's inverted indexing, and FAISS's hierarchical indexing.
- Indexing techniques, such as HNSW, IVF, LSH, and PQ, significantly impact search performance and accuracy.
- The choice of vector database depends on the specific use case, dataset size, and search requirements.

## ***Technical Details***

### #### Vector Database Architecture

Vector databases are designed to handle high-dimensional vector data, which requires specialized indexing techniques to enable efficient similarity search and querying. The architecture of a vector database typically consists of the following components:

- **\*\*Vector Storage\*\***: Stores high-dimensional vector embeddings, which are dense numeric representations produced by ML models.

- **Indexing**: Uses indexing techniques, such as HNSW, IVF, LSH, and PQ, to optimize query performance.
- **Query Processing**: Handles queries, such as similarity search and filtering, using the indexed vector data.

#### Indexing Techniques

Indexing techniques are crucial in vector databases, as they enable efficient similarity search and querying. The following are some common indexing techniques used in vector databases:

| Indexing Technique                        | Description                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| ---                                       | ---                                                                                                                                 |
| HNSW (Hierarchical Navigable Small World) | Graph-based indexing method that offers scalability advantages                                                                      |
| IVF (Inverted File)                       | Indexing technique that uses a combination of clustering and quantization to reduce dimensionality                                  |
| LSH (Locality-Sensitive Hashing)          | Hashing-based indexing technique that maps high-dimensional vectors to a lower-dimensional space                                    |
| PQ (Product Quantization)                 | Quantization-based indexing technique that reduces dimensionality by representing vectors as a product of lower-dimensional vectors |
| SQ (Simple Quantization)                  | Simple quantization-based indexing technique that reduces dimensionality by representing vectors as a set of discrete values        |
| RP (Random Projection)                    | Dimensionality reduction technique that projects high-dimensional vectors onto a lower-dimensional space using random projections   |

**Best Practices**

- **Choose the right indexing technique**: The choice of indexing technique depends on the specific use case, dataset size, and search requirements.
- **Consider scalability and efficiency**: Vector databases should be designed to handle large-scale datasets and provide efficient search performance.
- **Balance accuracy and efficiency**: The choice of indexing technique should balance search accuracy and efficiency trade-offs.
- **Use hierarchical structures**: For large datasets, consider using hierarchical indexing methods, such as HNSW, to improve search performance and accuracy.

**Conclusion & Future Work**

Vector databases are a new type of database designed to handle high-dimensional vector data, which is essential for various applications, including semantic search, recommendations, and natural language processing. This report provides a comprehensive review of vector databases, including their core functionalities, architectural differences, indexing techniques, and best practices. As the field of vector databases continues to evolve, future work should focus on improving the scalability, efficiency, and accuracy of vector databases, as well as exploring new applications and use cases.

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